

# Python Tsunami

Python I - An Introduction to Python

**Who are we?**

# Center for Health Data Science

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Center for Health Data Science (HeaDS)

Section for Health Data Science and Artificial Intelligence (HDSAI)

The mission of HeaDS is to strengthen health data science research at SUND:

- Visible hub for Health Data Science
- Research within the field of HDS
- SUND Data Lab



<https://heads.ku.dk>



## COURSES & WORKSHOPS

Data Science skills, Tools  
and HDS Topics

## COMMISSIONED RESEARCH

Commissioned Data Science Analysis  
Commissioned Supervision



## CONSULTATIONS

Need guidance? Drop by for a  
consultation on your research  
project

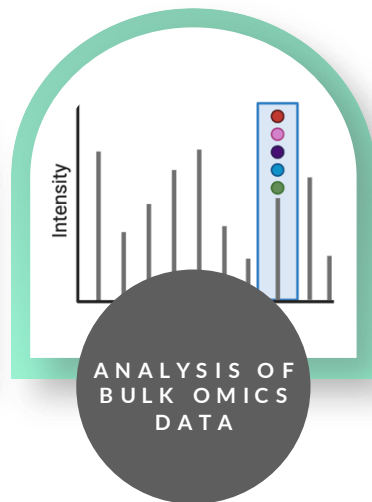
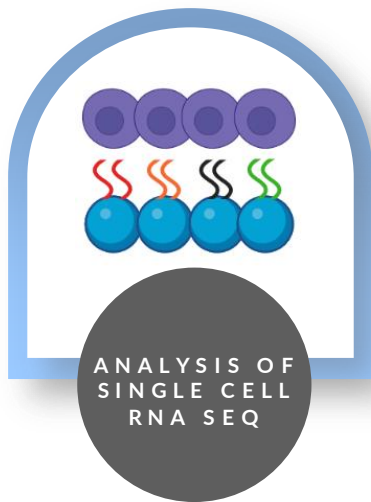
## MATCHMAKING

Conference, seminars and  
networking events - Join us!



# Commissioned Research

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- Machine Learning
- Biometric Data
- Metagenomics
- Epigenetics
- Biological Networks
- Seq. Motif Discovery
- Text mining
- *Biostatistical Models*

# HeaDS Courses



## Skill Sets at Different Levels:

- Programming & Data Science (R, Python)
- Biomedical Data Analysis
- Advanced Computational Workflows

From  
Excel to R



Python I



R for  
Data Science



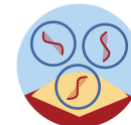
Python II for  
Data Science



Genomics  
Sandbox



RNA-seq  
Sandbox



HPC-Launch



HPC-Pipes



Practical AI



# The Python Tsunami Team

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## The Team:

Thilde Terkelsen

Suze Roostee

Iñigo Prada

Annike Kjølby Kristensen

Astrid Filt



### **Original Content:**

Henrike Zschach (Research IT)

Alberto Delgado (DTU, Biosustain),

Henry Webel (DTU, Biosustain),

Annelaura Nielsen (University Hospital),

Rita Colaço (PRI)

# About this course



| Day 1       |                                    | Day 2       |                                            | Day 3       |                         |
|-------------|------------------------------------|-------------|--------------------------------------------|-------------|-------------------------|
| 08:45-09:00 | Morning Coffee                     | 08:45-9:00  | Morning Coffee                             | 08:45-09:00 | Morning Coffee          |
| 09:00-09:30 | Intro to Python & Colab            | 09:00-09:30 | Functions                                  | 09:00-09:30 | Pandas recap quiz + Q&A |
| 09:30-10:10 | Variables & Data Types             | 10:00-10:30 | Basics recap quiz + Q&A                    | 09:30-10:30 | Visualization           |
| 10:10-10:20 | Break                              | 10:00-10:30 | Pandas: Series, Dataframes, Renaming       |             |                         |
| 10:20-11:00 | Iterables I: Lists                 | 10:30-10:45 | Break                                      | 10:30-10:45 | Break                   |
| 11:00-11:10 | Break                              | 10:45-11:30 | Pandas: Index, Select and Summarise        | 10:45-12:00 | Visualization           |
| 11:10-12:00 | Iterables II: Sets, Dicts & Tuples | 11:30-12:00 | Pandas: Modifying Data                     |             |                         |
| 12:00-13:00 | Lunch                              | 12:00-13:00 | Lunch                                      | 12:00-13:00 | Lunch                   |
| 13:00-14:15 | Booleans, Operators & Conditions   | 13:00-14:15 | Pandas: Groupby, Sort & Impute             | 13:00-13:30 | Local Installations     |
| 14:15-14:30 | Break                              | 14:15-14:30 | Break                                      | 13:30-16:00 | Dataset Exercise        |
| 14:30-16:00 | Loops                              | 14:30-15:15 | Pandas: Handling missing data + Pandas Q&A |             |                         |
| 16:00       | See you tomorrow!                  | 15:15       | See you tomorrow!                          | 16:00       | Course Finished!        |



# Python

# Why Python?

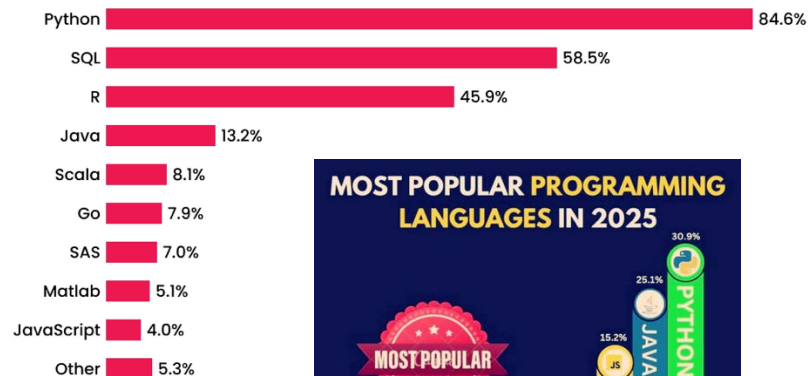


Python is a great programming language for both beginners and advanced programmers:

- Easy to grasp, close to natural language
- Many learning resources available
- Large community
- Libraries / Packages
- Makes machine learning 'easy'

## Required Programming Languages

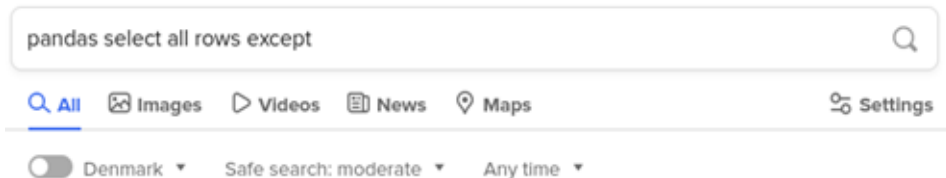
### Data Scientists 2025



# Where to get help



Online communities such as **stackoverflow** are important tools in programming.

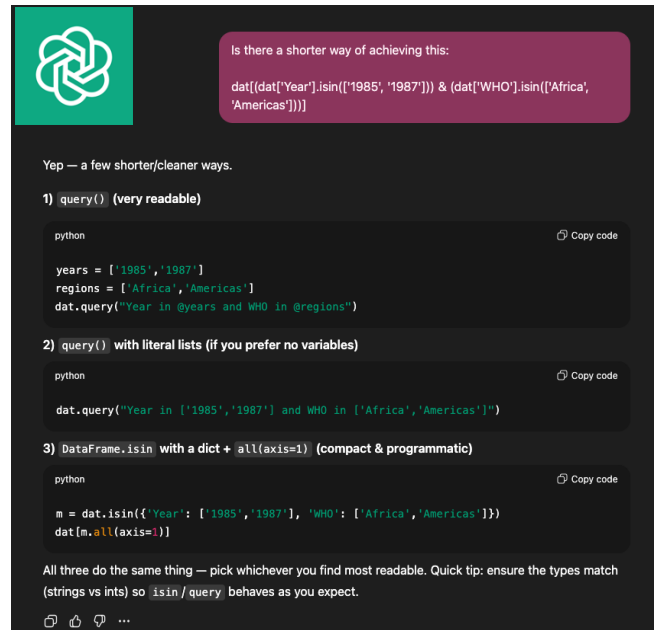


<https://stackoverflow.com/questions/28256761/select-pandas-rows-by-excluding-index-num...>

## select pandas rows by excluding index number - Stack Overflow

I'm looking to slice a **Pandas** dataframe by using index numbers. I have a list/core index with the index numbers that i do NOT need, shown below **pandas**. Stack Overflow ... Use a list of values to **select rows** from a **Pandas** dataframe. 2015. Delete a column from a **Pandas** DataFrame. 1290. How to drop **rows** of **Pandas** DataFrame whose value in a certain ...

Or ask an **AI tool**. But remember: “With great power comes great responsibility!”



# Python Environments



**Google Colab** is a Jupyter Notebook hosted on Google's servers, not your own machine. It still runs in your browser.

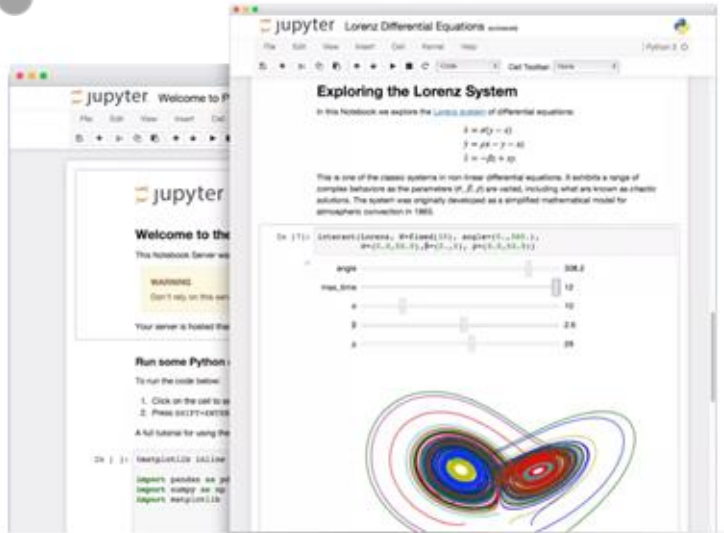
- Tool to write, execute and share Python code through the browser
- Requires no setup to use and provides free access to computing resources on Google's servers including GPUs
- Connected to a Google account and data and notebooks can be accessed through Google Drive.

**We'll use Colab during the course.**

# Jupyter notebook



Jupyter Notebook is an **open-source application** to create and share documents that contain code, visualizations and text (markdown).



- Browser-based development environment for creating, running and sharing Python code
- Combine code with text and output – generates; html, pdf, website
- Runs from your **local installation**. I.e. you need Python and the libraries you want to use installed on your computer



Text cell ↓


Close

A variable is simply a name that holds a value.

Variable containers --> name and value

- Create variables by assigning a value to a name (just like using variables in math).

- Variable names should be meaningful, i.e. not just `a`, `b`, `c`

- Variables are always **assigned** with the variable name on the left and the value on the right of the ***equals*** sign. For instance:

- `a_variable = 100`

- assigned to other variables: `another_variable = a_variable`

- reassigned at any time: `a_variable = 435`

- assigning several variables at the same time: `all, at, once = 1, 130, 43`

- Variables **must** be assigned before they can be used.

```
# What is the container and what are the data here?
```

Code cell 



# Jupyter notebook



Getting help  
inside the  
notebook

```
[5] ✓ Os help(print)
... Help on built-in function print in module builtins:

print(*args, sep=' ', end='\n', file=None, flush=False)
    Prints the values to a stream, or to sys.stdout by default.

    sep
        string inserted between values, default a space.
    end
        string appended after the last value, default a newline.
    file
        a file-like object (stream); defaults to the current sys.stdout.
    flush
        whether to forcibly flush the stream.
```

## Settings

|               |   |                                                                                                 |
|---------------|---|-------------------------------------------------------------------------------------------------|
| Site          | > | <input checked="" type="checkbox"/> Show AI-powered inline completions                          |
| Editor        | > | <input type="checkbox"/> Consented to use generative AI features                                |
| AI assistance | > | <input type="checkbox"/> Hide generative AI features                                            |
| Colab Pro     | > | Gemini can make mistakes, so double-check responses and <a href="#">use code with caution</a> . |
| GitHub        | > |                                                                                                 |
| Miscellaneous | > |                                                                                                 |

If you don't like  
AI to do the  
work for you

(Colab specific)

# Python Libraries

Python has many libraries, also called packages, that other programmers have developed. Find and **use** them!

Well-maintained libraries generally are:

- Tested
- Optimized
- Documented

During this course we will use:

- Pandas (all the data analysis)
- Math (basic math)
- Seaborn (visualization)



Running Python from a local installation requires you to have libraries **downloaded & installed** before you can use them.

On **Google Colab** you can generally just import, they are already installed.

- Import the math library:

```
import math
```

- Now I can use functions from that library, i.e. calculating the logarithm or square root:

```
math.log(3)
```

```
math.sqrt(4)
```

# Course Material



Center-for-Health-Data-Science / PythonTsunami Public

forked from [pythontsunami/teaching](#)

[Code](#) [Issues](#) 10 [Pull requests](#) [Discussions](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

spring2022 16 branches 0 tags

[Go to file](#) [Add file](#) [Code](#)

This branch is 111 commits ahead of [pythontsunami/teaching:heads](#). [Contribute](#) [Fetch upstream](#)

**hezscha** Add files via upload 078c67a 2 days ago 376 commits

|                        |                                  |            |
|------------------------|----------------------------------|------------|
| Conditionals           | Add files via upload             | 6 days ago |
| Exercise               | Add files via upload             | 2 days ago |
| Functions              | Add files via upload             | 2 days ago |
| Introduction_and_tools | Add files via upload             | 7 days ago |
| Iterables              | Add files via upload             | 2 days ago |
| Loops                  | Add files via upload             | 2 days ago |
| Pandas                 | Minor changes to pandas examples | 6 days ago |
| Recap                  | Add files via upload             | 9 days ago |
| Variables_data_types   | Add files via upload             | 6 days ago |

You can find the course material here:

<https://github.com/Center-for-Health-Data-Science/PythonTsunami>

# Course Material



spring2022 PythonTsunami / Variables\_data\_types / [Go to file](#) [Add file](#) [...](#)

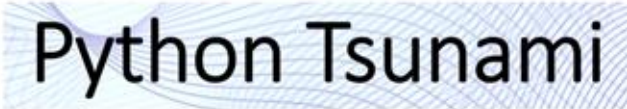
This branch is 111 commits ahead of pythontsunami/teaching:heads. [Contribute](#) [Fetch upstream](#)

**hezscha** Add files via upload ab1c127 6 days ago [History](#)

..

|                           |                                       |            |
|---------------------------|---------------------------------------|------------|
| README.md                 | Minor fixes                           | 7 days ago |
| variables.ipynb           | Update Colab link within the notebook | 6 days ago |
| variables_solutions.ipynb | Add files via upload                  | 6 days ago |

**README.md**

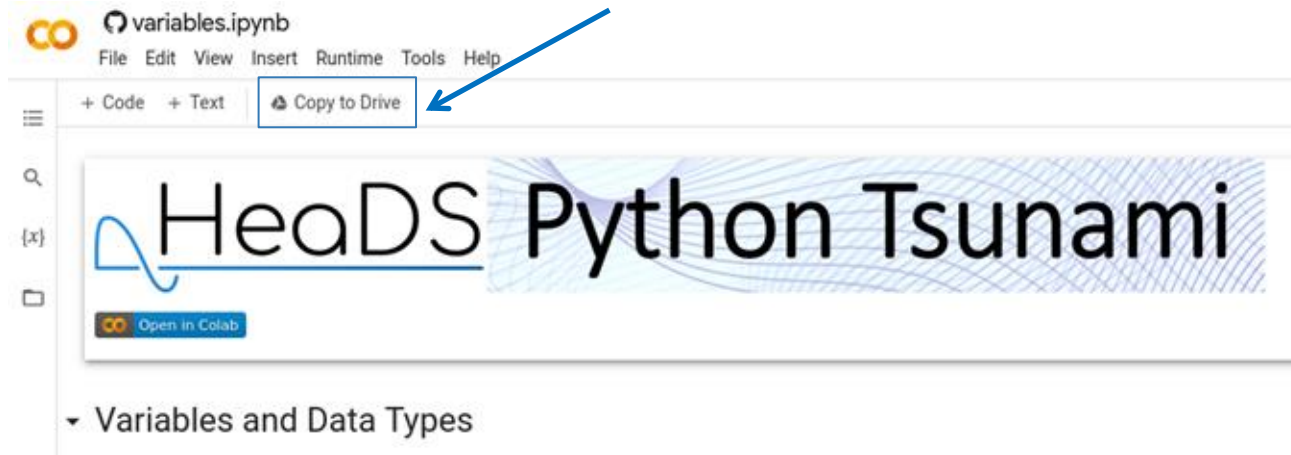
| notebook                                                      | content                  |
|---------------------------------------------------------------|--------------------------|
| <a href="#">variables.ipynb</a> <a href="#">Open in Colab</a> | Variables and data types |



# Course Material



Remember to **save a copy** to your own google drive so you can save your notes and exercises!



# Short Introduction

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Take the next 5 mins to introduce yourself at your table:

- Name
- Position/Unit
- What you do (very briefly!)
- Why would you like to learn Python?